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Combining Preferred Activities With Peer Support to Increase Social Interactions Between Preschoolers With ASD and Typically Developing Peers

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Abstract

This study evaluated the effects of using preferred activities with peer support on the social interactions of preschoolers with autism spectrum disorder (ASD) and their typically developing peers. Two preschool-aged children with ASD and six peers in an inclusive classroom participated in this study. A multiple-probe single case experimental design across three peers was employed. Results indicated that the participants' social initiations and responses, as well as the percentage of time engaged in social play, increased following the preferred activity. Gains were also demonstrated across social interactions and time engaged in the preferred activity plus peer support condition. Increases in social interaction behaviors in typically developing peers were also observed, and positive intervention gains were maintained during 5-week follow-up sessions. Implications for research and practice for improving interaction between children with and without ASD in inclusive preschool settings are discussed.

Keywords

autism spectrum disorder, China, inclusion, preferred activities, peer support, social interactions

Autism spectrum disorder (ASD) is a neurodevelopmental disorder characterized by social communication impairments and restricted interests and repetitive behaviors (American Psychiatric Association, 2013). One of the most obvious challenges associated with ASD is difficulties with peer social interaction, including initiating, responding to, and maintaining interaction with others in social settings (Koegel et al., 2008). These social challenges can adversely affect interactions between children with ASD and their peers, rendering inclusion in school settings difficult and challenging for practitioners. Thus, it is necessary to develop social skills interventions that can be successfully implemented in inclusive classroom settings to facilitate appropriate social interactions between children with ASD and their classmates.

Research suggests that social skill interventions for children with ASD are most effective when implemented in the natural setting within the context of typical routines and activities, and numerous studies have demonstrated that the use of naturalistic behavioral strategies, often incorporating the preferred interests or activities of the child with ASD, can improve social skills such as joint attention, interactive play, peer interaction, and social engagement for children with ASD in inclusive settings (Koegel et al., 2008). Watkins et al. (2019), for example, developed an intervention package incorporating preferred interest-based activities that resulted in increased social interaction between children with ASD and their typically developing classmates. After establishing the preferred interests and activities of four preschoolers with ASD, the researchers incorporated those interests into play activities mutually appealing to their peers. The intervention involved brief sessions of adult instruction, modeling, and response to child questions during the interest-based activity, and results indicated that social interaction behaviors were higher in intervention conditions compared with business as usual free play sessions. Incorporation of preferred activities into

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play sessions demonstrated a positive effect on social interaction; however, such interactions decreased to a low level when the intervention was withdrawn, suggesting that social interaction remains challenging for children with ASD in inclusive settings without additional support. Investigation of additional strategies that will help support and maintain social interaction between children with ASD and their classmates when using this intervention is therefore needed.

Peer-mediated or peer support strategies have demonstrated great effectiveness in improving social behavior and supporting the maintenance of social skills for children with ASD (Chan et al., 2009; Watkins et al., 2015). Peer support strategies are considered an evidence-based practice and have been used successfully in inclusive classroom settings to increase social interactions between children with ASD and their typically developing peers (Watkins et al., 2015). These strategies are often incorporated within multicomponent behavioral intervention packages and may help increase generalization of skills in natural settings for children with ASD (Chan et al., 2009). Effective strategies used in peer support interventions commonly include modeling, prompting, and positive reinforcement of appropriate social behaviors (Watkins et al., 2015). In addition to improving the social interactions of children with ASD, research has demonstrated that typically developing peers engage in social interactions much more frequently when taught to use peer support strategies (Gonzalez-Lopez & Kamps, 1997; McGrath et al., 2003). For example, Hart and Banda (2018) trained four peer groups to implement peer support strategies. For each child with ASD, three typically developing peers formed one peer group. Results indicated clear increases in peer responses to social initiations made by the child with ASD. Owen-Deschryver et al. (2008) also examined the use of peer support strategies for school-aged children with ASD and their typically developing peers. This study found peer initiations toward students with ASD increased dramatically. However, the social responses made by typically developing peers were not examined and follow-up sessions were not conducted. Considering the optimal outcome of social interaction interventions is long-lasting change for both typically developing peers and children with ASD, it is necessary to examine long-term follow-up sessions to evaluate the durability of intervention results, especially in inclusive settings. Furthermore, research is needed that considers additional outcomes important to social interaction, such as time spent engaged in social play with a peer. Examining these multiple outcomes will provide a more nuanced analysis of the effects of these strategies in the classroom.

Along with increases in social interaction, research has also demonstrated that peer support strategies can help reduce inappropriate social interactions of children with ASD. Gonzalez-Lopez and Kamps (1997) found that following peer training, inappropriate behaviors (i.e., grabbing, disruptions, leaving the group) of four children with ASD decreased. Moreover, in a peer-mediated pivotal response training by Brock and colleagues (2018), inappropriate play (i.e., breaking play rules or stereotypic behaviors) and social behaviors (i.e., not responding to peers) decreased to low levels. However, these studies also lacked maintenance sessions to evaluate the long-term effects of peer support on the reduction of inappropriate social interactions, so additional questions regarding the durability of intervention results remain.

Much of the research in this area has been conducted in the United States, yet the inclusion of students with ASD in regular educational settings is a recommended and growing practice worldwide (Watkins, Ledbetter-Cho, et al., 2019). In China, access to and participation in general education for Chinese students with ASD varies greatly by geographic area, disability severity, resources, and the availability and knowledge of professionals (Hu et al., 2018). With the inclusive education reform in China, more and more students with ASD are enrolled in regular elementary schools (An et al., 2018). In large cities (e.g., Beijing), inclusive preschool classrooms often have one special education teacher and three general education teachers, with around 33 typically developing children and two or three children with disabilities, for a student–teacher ratio of 9:1 (Hu et al., 2018). Thus, the development of effective interventions for students with ASD that are feasible to implement in this specific context is a pressing research priority. Therefore, the purpose of this study was to evaluate the effects of an intervention package on social interactions of preschoolers with ASD in an inclusive Chinese classroom. Specifically, we replicated the Watkins, O'Reilly, et al. (2019) study using preferred activities, but extended it by including peer support strategies within the intervention package. We sought to assess whether the intervention package would (a) increase interaction, including social initiations, social responses, and the time engaged in social play for children with ASD; (b) increase interaction, including social initiations and social responses, of six typically developing peers; (c) have an effect on the inappropriate social interactions of children with ASD and typically developing peers; and (d) be socially acceptable to stakeholders in an inclusive Chinese classroom.

Method

Participants

Institutional review board approval and consent from teachers, children, and parents were obtained for this study. Yaoyao (pseudonym), a 6-year-old Chinese girl diagnosed with ASD, and three typically developing classmates (Peer 1, female, 6 years; Peer 2, female, 6 years; Peer 3,

male, 5.5 years) participated in this study. Yaoyao's IQ score was 78 and her verbal skills, assessed with the Chinese version of the Verbal Behavior Milestones Assessment and Placement Program (VB-MAPP; Huang & Li, 2017; Sundberg, 2008), were at Level III (30–48 months). She used full sentences, made eye contact with others to request preferred items, and could answer basic social questions. She exhibited grabbing behaviors when her requested items were not provided immediately. During group play activities, Yaoyao could observe peers with verbal prompts from the teacher. However, Yaoyao seldom asked questions to teachers or peers. Yaoyao was observed to spend most of her free playtime in the play area, but she did not interact with peers and engaged in solitary play or delayed echolalia. She had functional play skills and could engage with play items such as toy sets in the manner in which they were intended.

Baobao (pseudonym), a 5-year-old Chinese boy diagnosed with ASD and three typically developing classmates (Peer 4, male, 5 years; Peer 5, male, 5 years; Peer 6, male, 5.5 years) participated this study. Baobao's IQ score was 85, and he scored at Level III on the VB-MAPP. Baobao generally only spoke to teachers to request books or toys. He could ask for items with full sentences when prompted, "What do you want?" Baobao could label at least 500 common objects; however, he only initiated communication when requesting preferred items and seldom made eye contact with others. He rarely introduced himself to others, invited peers to join activities, or followed rules when playing games with others. Observations of his play indicated that he preferred solitary play and rarely engaged in play with typically developing peers even though he was seated with them. He spent most of his time playing with blocks and train toys. He could request with full sentences and respond to two-step directions given by teachers but usually needed prompts to complete tasks.

We asked the head teachers to nominate typically developing peers in their classrooms to participate in this study. To meet criteria as a peer play partner, the child had to (a) demonstrate typical verbal communication skills, (b) demonstrate appropriate social skills, (c) have regular school attendance, (d) be in the same classroom as the child with ASD, (e) be willing to serve as a peer and play with the participant with the preferred activity materials (i.e., kitchen set and train set described below), and (f) show an interest in interacting with the child with ASD. We included two girls and four boys, aged 4 to 6, as peers in this study.

Setting and Materials

The study took place in two inclusive preschools in Beijing and Fuzhou, China. Yaoyao's class in Beijing had four children with ASD or other developmental delays, 20 typically developing children, two lead teachers, and two teaching assistants. The special education teacher had a degree in special education, and the general education teacher had a degree in early childhood education. The children with disabilities attended the inclusive classroom for 8 hr a day and received specialized instruction in the resource room for 1 hr daily. Baobao's class in Fuzhou had two children with ASD and 28 typically developing children, three lead teachers, and one teaching assistant. The special education teacher had a degree in special education, and the two general education teachers had a degree in early childhood education. The children with ASD attended the inclusive classroom for 8 hr a day without specialized pull-out instruction. For both Yaoyao and Baobao, all sessions were conducted in the free play area of their respective inclusive classrooms, and all materials in the study involved play items typically found in the classroom setting. Teachers provided peers with a weekly certificate of "Helpful Star" after intervention sessions.

Operational Definitions and Data Collection

We measured the number of social initiations and social responses between participants and peers. A social initiation was defined as any verbal or nonverbal behavior directed to another child without any preceding verbal or nonverbal behavior (e.g., saying, "Look what I made!" or showing an item to another child). A single social initiation was recorded if, for example, the child initiated both verbally and with gestures in the same instance (e.g., the child said, "Look, I made cookies. Do you want one?" while offering a plate to the other child). A social response was defined as any verbal or nonverbal behavior in response to a preceding verbal or nonverbal initiation by another child within 5 s of that initiation (e.g., looking at the item shown by another or commenting, "It looks interesting."). A single social response was recorded if, for example, the child responded both verbally and with gestures in the same instance (e.g., following an initiation of "Look, I made cookies. Do you want one?" from the other child, the child responded verbally, saying "Sure, yum!" while making an eating gesture.)

An inappropriate social interaction was defined as any verbal or nonverbal behavior directed to another person that was inappropriate in the context, such as grabbing toys from others without permission or yelling at others. In addition, an inappropriate social interaction was scored when a child did not emit a verbal or nonverbal response to a preceding initiation from another child that had provided a specific opportunity for a response. For example, an inappropriate interaction was recorded if the participant did not respond to the peer with speech or with gestures after being asked by the peer, "Do you want this red train?"

Engagement in social play was measured using 5-s whole interval recording. An interval of engagement in social play was scored if both children were engaged in play

activities during the whole 5-s interval while also interacting together (i.e., either child emitted an initiation and/or response behavior as described above) within the 5-s interval. If there were no initiation or response behaviors exhibited while playing together, engagement in social play was not scored for the interval.

Design and Procedures

We used a multiple-probe-across-peers with two intervention phases design (i.e., A-B-A-B+C) to examine the effects of the two intervention procedures (i.e., preferred interest activity and preferred interest activity with peer supports) on social interactions between the target children and their peers. The researcher collected intermittent probes during baseline and continuous daily probes during Intervention 1 and Intervention 2.

We conducted Baseline 1 and Baseline 2 sessions in a typical free play setting. During the 15-min baseline sessions, the participant with ASD and peer were directed to the play area of the classroom and told it was time to play. As in Watkins, O'Reilly, et al. (2019), the participant and peer were allowed to engage with any of the items typically available during playtime, and no adult prompts or reinforcement of any social behaviors were provided to the children. Adult instruction was provided only if a child attempted to leave the play area of the classroom, in which case the researcher redirected him or her back to the play area, or in cases when inappropriate classroom behavior was displayed, in which case the researcher reminded the child of classroom expectations and redirected the behavior with verbal instruction. Items related to the preferred play activities determined by the preference assessment described below were present during baseline but not directly embedded into a specific play activity with the peer, nor were the children instructed to use certain play items.

Prior to beginning Intervention 1, we conducted preference assessments to establish the play activities most preferred by the participants that could be incorporated into mutually enjoyable play activities with peers. To select Yaoyao's preferred activities, we conducted a free operant assessment (Roane et al., 1998). We chose this preference assessment as it is appropriate for children like Yaoyao who displayed challenging behavior (i.e., grabbing) when presented with an array of tangible reinforcers. The free operant assessment took 20 min. Researchers provided 10 toys and observed the duration of time that Yaoyao played with each toy. These play materials included a toy kitchen set with a cooking range, sink, cookware, utensils, and food items. The kitchen set was selected as Yaoyao's preferred play activity that was also mutually enjoyable to her peers. To select Baobao's preferred interests, we used a multiple-stimulus-without-replacement to assess 10 potential play activities (DeLeon & Iwata, 1996). These play materials included engines, train cars, train tracks, stations, and railway items. We used multiple-stimulus-without-replacement as it is appropriate for children like Baobao who can choose from an array of potential items in a short amount of time without displaying challenging behaviors. We conducted three assessments and Baobao chose a train set as the first item to play with in these assessments. Before the start of each session in Interventions 1 and 2, the researcher asked the participants' preferences for toys to play with. The participants consistently chose the preferred activity materials identified in the preference assessments, and these materials remained the same for the two intervention phases.

Intervention 1 consisted of the play activity which incorporated the preferred activity of the child with ASD and directly replicated the procedures described in Watkins, O'Reilly, et al. (2019). The participant and peer were again directed to the play area of the classroom, and the researcher introduced the preferred activity (e.g., "Today we're going to bake a birthday cake!"). The researcher then provided brief verbal instruction of the preferred activity that the participant and peer would be engaging in during the play session, followed by modeling of the activity. During the 15-min play session, the researcher was available to respond to children's questions that arose as it pertained to the preferred activity (e.g., helping to connect a piece of the railway if asked), but as in baseline, no adult prompts or reinforcement were provided to either the participant or peer for any social interaction behaviors. If needed, the researcher verbally redirected the child back to the play area or redirected inappropriate behavior as in baseline. The average number of play sessions of Intervention 1 was 6.3 (range = 5–8).

Intervention 2 consisted of the same preferred activity intervention as described above with the addition of peer support strategies adapted from Hu et al. (2018). We conducted a preintervention peer training session (20–30 min) with each peer before entering the second intervention condition. During peer training, the instructor used verbal instructions and modeled social initiations and responses, each type of prompt strategy, and various forms of reinforcement (Hu et al., 2018). The peer reached mastery when each target skill was demonstrated three times through role-play with the instructor. The training session included training peers to help participants label items, prompting participants to use social responses and initiations, and providing positive reinforcement to the participants for demonstrating social and play behaviors. See Table 1 for content, target skills, instruction, and examples of the peer training session.

The instructor conducted a brief review session (3–5 min) with the peer immediately before each Intervention 2 session. Each intervention 2 session lasted 15 min. Each session began with the instructor suggesting a play theme

Table 1. Content, Target Skills, Instruction, and Examples of Preintervention Peer Training.

Content	Target skill	Instruction	Examples
Preferred activities	Naming items	Labeling name, function, or features of toy items	"Give me a brown square plate." "Please hand me a small cup."
Social skills	Response	Eye contact, answering questions, providing help, responding to comments and requests.	Looking at the item pointed by the child. Getting a plate when asked by the child.
	Initiation	Calling peer's name looking at peer, asking questions, making requests or comments, initiating joint attention, seeking help.	Asking, "Can you hold this? " Say, "Look" when showing an item
Intervention strategies	Prompt	Verbal instructions, echoic prompts, gestural prompts, physical assistance	"You can put the veggies in the pot and cook them."
	Reinforcement	Praise, clapping hands, cheering	"You are doing great!" "Thanks for your help!"

related to Yaoyao's preferred kitchen set (e.g., "It's dinner time! Let's cook something yummy!") or Baobao's preferred train set (e.g., "We are going to take the train to the farm!") to engage the two children in the play activity. During the play session, the instructor monitored the children to ensure safety and did not provide any guidance on increasing interactions. The instructor only provided assistance if the children had any conflict and did not provide any instruction for the children during the session. Immediately after the session, the instructor provided specific feedback to the peer regrading social interactions, prompts, and reinforcement used by the peer during session (e.g., "It was great that you responded to Yaoyao by holding the plate when she asked you to," or "When you made salad, you did it all by yourself. You can try asking Yaoayo to help next time."). As in baseline and Intervention 1 sessions, if a child left the play area, researchers redirected him or her back, and if participants or peers demonstrated inappropriate behaviors, the instructor reminded the child of classroom expectations (e.g., "Do not grab other's toys. Hands to self.") and redirected the participant with verbal instructions (e.g., "Look at the stove and play with it"). The average number of play sessions of Intervention 2 was 5.3 (range = 5–8). Follow-up sessions were conducted in the same manner as Intervention 2 sessions, except no review and feedback for the peer was provided before and after each session. We conducted follow-up sessions at Weeks 1, 3, and 5 following Intervention 2.

Data Analysis

In addition to visual inspection of the data to determine intervention effectiveness, we quantitatively analyzed the data using Tau-*U*, a nonparametric measure appropriate for single case designs to examine within-phase trend and across-phase differences (Parker et al., 2011). Tau-*U* is directly interpreted as one continuous indicator of intervention effect. Olive and Smith (2005) suggested comparing the first baseline and the last implemented intervention to identify the effect size in multiple treatment design. Therefore, in this study the data of the first baseline and the Intervention 2 were compared to calculate the Tau-*U* across three peers. An effect size of 0.20 indicates a small change or effect, 0.20 to 0.60 indicates a moderate effect, 0.60 to 0.80 indicates a large effect, and above 0.80 indicates a very large effect (Vannest & Ninci, 2015).

Procedural Fidelity, Interobserver Agreement, and Social Validity

Researchers trained two second-year master's students before conducting interobserver agreement (IOA) and procedural fidelity by providing video recordings of each participant and peer. Training included recording occurrences of social interactions, inappropriate social interactions, and engagement in social play. The records of these two students were compared and disagreements during training were resolved by reviewing the video clips until agreement was reached. The training mastery criterion was IOA of 95% for three consecutive training sessions. During the IOA check, if agreement was lower than 90%, the researcher examined the video for further analysis. Procedural fidelity was calculated by dividing the number of steps implemented accurately by the total number of steps and multiplying by 100. See Table 2 for the checklist used to assess procedural fidelity. Procedural fidelity and IOA were assessed for at least 30% of all conditions for the three peers. The procedural fidelity averaged at 99.2 across all sessions (range = 96.7%–100%). Point-by-point IOA was assessed by dividing the number of agreements by the total agreements and disagreements and multiplying by 100. Kappa reliability coefficients were obtained using IBM SSPS Statistics Version 19. IOA averaged 90.43% (range = 81.5%–100%; k = .97, range = .75–1) for Yaoyao's social initiations, 95.7% (range = 88.9%–100%; k = .98, range = .96–1) for her social responses across three peers, and 97.6% (range = 96%–100%; k = .99, range = .98–1) for inappropriate social interactions. The IOA for three

Table 2. Procedural Integrity Checklist.

Items	Yes/No
Baseline:	
1 Did the instructor tell the children it was free playtime in the beginning of each session?	☐ ☐
2 Were the play areas arranged in a way that all children could move freely and safely?	☐ ☐
3 Were the play materials in each play area placed at children's eye level and easily accessible to children?	☐ ☐
4 Did the instructor tell the children to clean up all items 5 min before the end of each session?	☐ ☐
5 Did the instructor monitor if children interacted with each other nicely and provide assistance if necessary?	☐ ☐
6 Did the session last for 15 min?	☐ ☐
Intervention 1 (Preferred activities):	
1 Did the instructor begin the session with a suggested play theme?	☐ ☐
2 Did the instructor present the materials for preferred activities to children?	☐ ☐
3 Did the instructor monitor if children interacted with each other nicely and provide assistance if necessary?	☐ ☐
4 Did the instructor tell the children to clean up all items 5 min before the end of each session?	☐ ☐
5 Did the session last for 15 min?	☐ ☐
Intervention 2 (Preferred activities + peer support)	
1 Did the experimenter review the strategies with the peer before the session?	☐ ☐
2 Did the instructor begin the session with a suggested play theme?	☐ ☐
3 Did the instructor present the materials for preferred activities to children?	☐ ☐
4 Did the instructor monitor if children interacted with each other nicely and provide assistance if necessary?	☐ ☐
5 Did the peers initiate interactions appropriately?	☐ ☐
6 Did the peers respond to interactions appropriately?	☐ ☐
7 Did peers provide prompts and reinforcement appropriately?	☐ ☐
8 Did the instructor tell the children to clean up all items 5 min before the end of each session?	☐ ☐
9 Did the session last for 15 min?	☐ ☐
Follow-up (same as Intervention 2 without Items 1 and 9)	☐ ☐

peers averaged at 94.1% (range = 85%–100%; k = .96, range = .79–1) for initiations, 95.1% (range = 84%–100%; k = .96; range = .89–1) for responses, and 96.3% (range = 95%–100%; k = .97, range = .96–.99) for inappropriate social interactions. IOA for engagement of social play ranged from 81.8% to 100% with an average of 94.4% (k = .93, range = .78–1). IOA averaged 95.6% (range = 92%–100%; k = .98, range = .95–1) for Baobao's social initiations, 98.5% (range = 96.5%–100%; k = .99, range = .96–1) for his social responses across three peers, and 95.5% (range = 95%–100%; k = .94, range = .92–.96) for inappropriate social interactions. The IOA for three peers averaged at 95.5% (range = 92%–100%; k = .96, range = .89–1) for initiations, 95.9% (range = 90%–100%; k = .93; range = .92–1) for responses, and 98.2% (range = 97%–100%; k = .95; range = .94–.97) for inappropriate social interactions. IOA for engagement of interactive play ranged from 93.9.8% to 100% with an average of 96.4% (k = .95, range = .92–1).

Teachers, parents, and peers completed a 14-item questionnaire on acceptability, effectiveness, and satisfaction related to this study. Items 1 to 5 pertained to intervention acceptability in terms of content, activities, needs, and cultural adaptation. Items 6 to 11 assessed intervention feasibility including time periods, instructors, schedule, locations, and financial burdens. Items 12 to 14 examined satisfaction in participants' social initiations, social responses, and communication. Each item was rated on a 5-point Likert-type scale (1 = *not satisfied or not helpful* to 5 = *very satisfied or very helpful*).

Results

Children With ASD

Figures 1 and 2 show the frequency of social initiations, social responses, and inappropriate social interactions for Yaoyao and Baobao and the typically developing peers across conditions. During the first baseline sessions, both Yaoyao and Baobao exhibited few social initiations or responses (range = 0–2 for both Yaoyao and Baobao) and low levels of inappropriate social interactions (i.e., nonresponse or grabbing behaviors) at baseline sessions across peers (range = 0–2 for Yaoyao and range = 0–3 for Baobao). Intervention 1 (i.e., the preferred activity with brief adult instruction, modeling, and response to child questions) led to increases in the level of social interactions between Yaoyao and the three typically developing peers (social initiation: $M_{\text{yaoyao-peer1}}$ = 13, range = 11–17; $M_{\text{yaoyao-peer2}}$ = 7.67, range = 6–10; $M_{\text{yaoyao-peer3}}$ = 10.0, range = 4–20; social

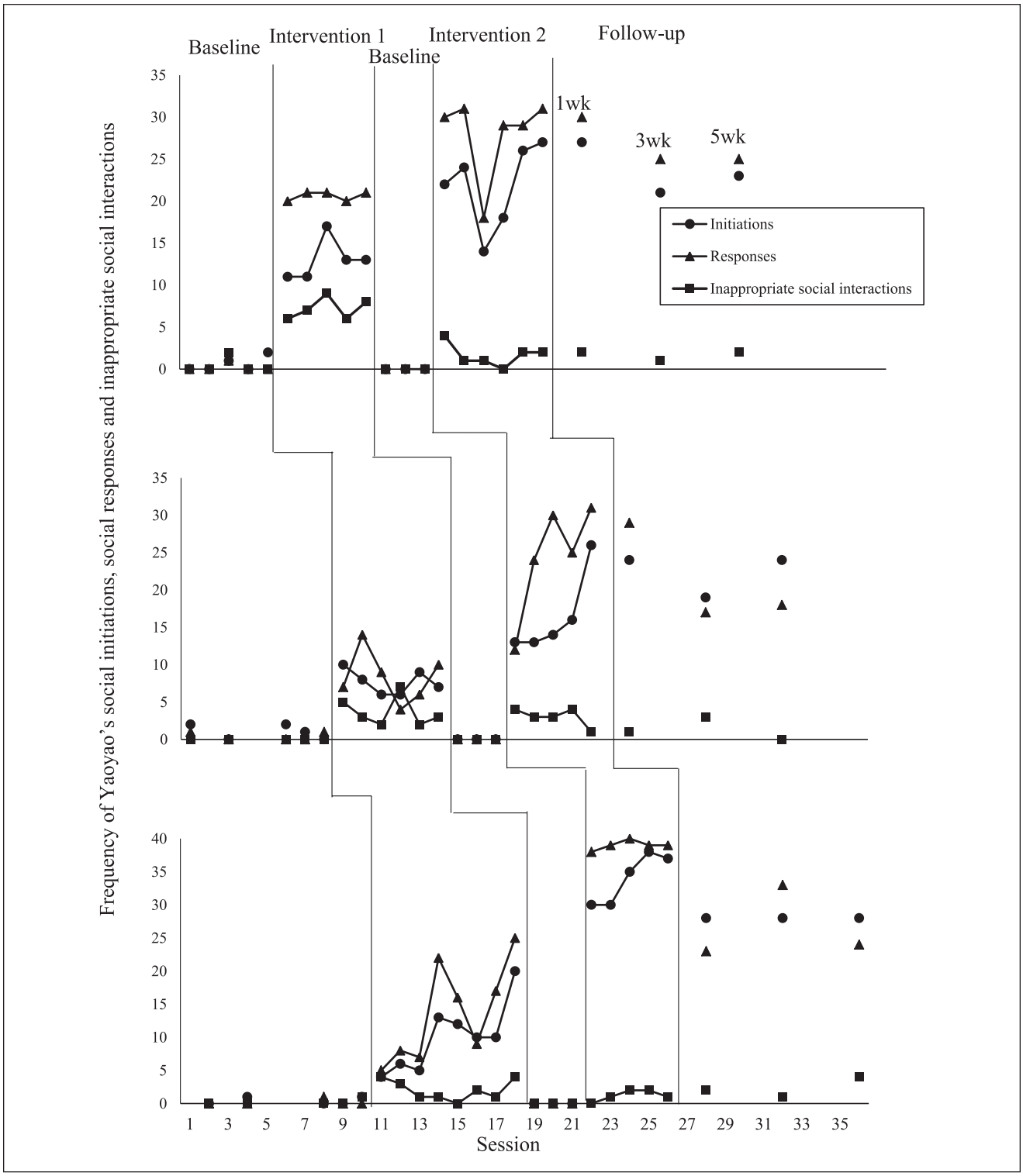


Figure 1. Frequency of Yaoyao's social initiations, social responses, and inappropriate social interactions directed to peer per play session.

response: $M_{\text{yaoyao-peer1}} = 20.6$, range = 20–21; $M_{\text{yaoyao-peer2}} = 8.33$, range = 4–14; $M_{\text{yaoyao-peer3}} = 13.6$, range = 5–25). Yaoyao's social responses displayed an ascending trend with variability. Tau-*U* scores fell within the highly effective range and suggested a strong treatment effect (Tau-$U = .81$; $p = .00$). Yaoyao's social initiations progressively increased (Tau-$U = .\ 0.84$: $p = .00$). The preferred play activities also resulted in a clear and rapid increase in

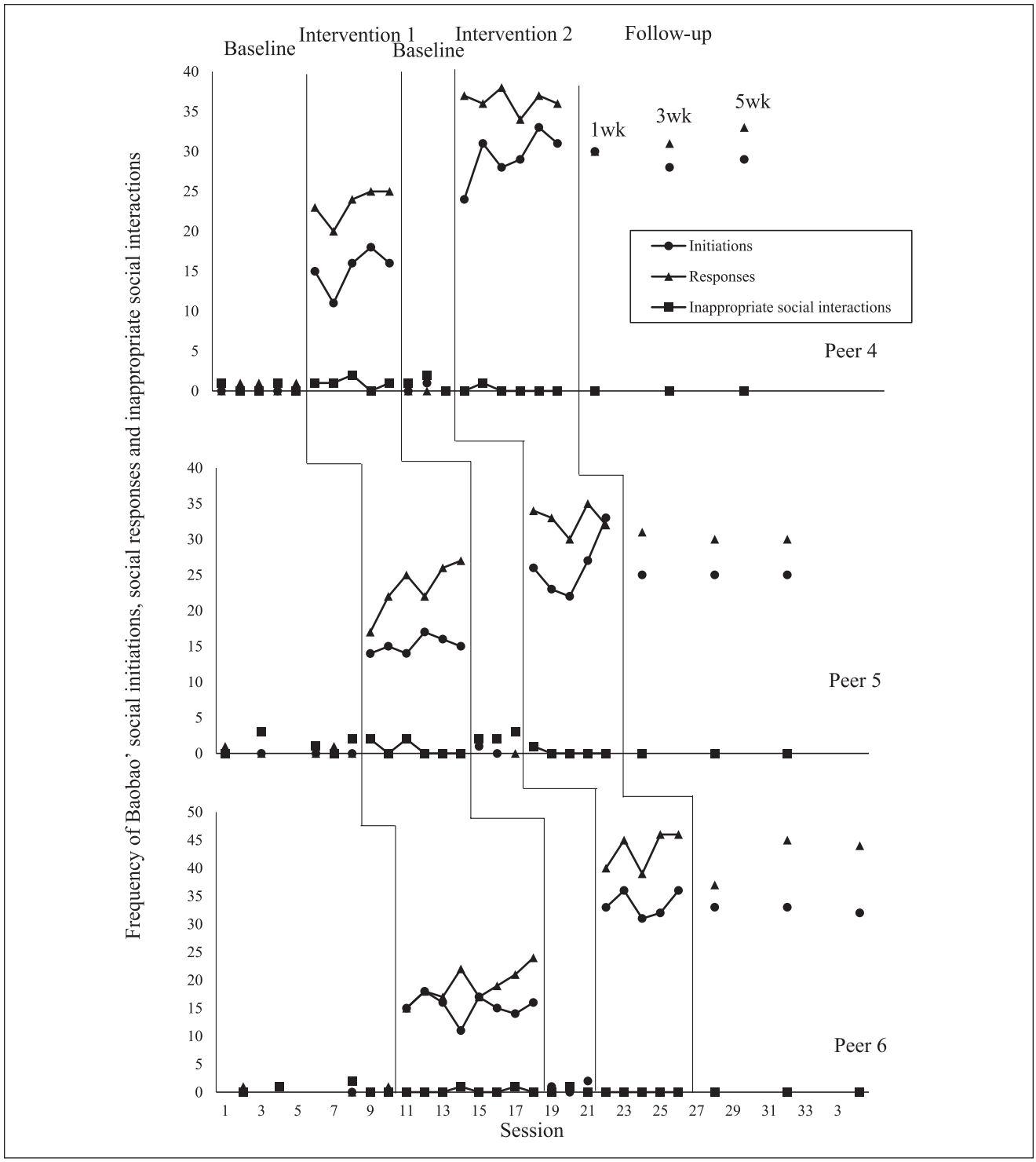


Figure 2. Frequency of Baobao's social initiations, social responses, and inappropriate social interactions directed to peer per play session.

the level of social initiations and social responses between Baobao and the other three typically developing peers (social initiation: $M_{\text{baobao-peer4}}$ = 15.2, range = 11–18; $M_{\text{baobao-peer5}}$ = 15.2, range = 14–17; $M_{\text{baobao-peer6}}$ = 15.3, range = 11–18, Tau-U = .91; p = .00; social response: $M_{\text{baobao-peer4}}$ = 23.4, range = 20–25; $M_{\text{baobao-peer5}}$ = 23.2, range = 17–27; $M_{\text{baobao-peer6}}$ = 19.1, range = 15–24, Tau-U = .95; p = .00). Although interaction increased, inappropriate social interactions also increased ($M_{\text{yaoyao-peer1}}$ = 7.2, range = 6–9; $M_{\text{yaoyao-peer2}}$ = 3.67, range = 2–7; $M_{\text{yaoyao-peer3}}$

= 2, range = 0–4). The most common inappropriate social interactions for Yaoyao included grabbing toys and not responding to social initiations delivered by peers. Baobao's inappropriate social interactions remained at a low level during the Intervention 1 sessions ($M_{\text{baobao-peer4}} = 1$, range = 0–2; $M_{\text{baobao-peer5}} = 0.67$, range = 0–2; $M_{\text{baobao-peer6}} = 0.33$, range = 0–1).

When we again reverted to baseline conditions, Yaoyao's social interactions and inappropriate social interaction decreased to zero among the three peers. Babao's social interaction with peers immediately decreased to low levels and inappropriate social interactions maintained at a low level. With the implementation of Intervention 2 (i.e., peer support programmed into the preferred activity intervention), social initiations and social responses between Yaoyao and her peers (social initiation: $M_{\text{yaoyao-peer1}} = 21.8$, range = 14–27; $M_{\text{yaoyao-peer2}} = 16.4$, range = 13–26; $M_{\text{yaoyao-peer3}} = 34$, range = 30–38, Tau-$U = .92$; $p = .00$; social response: $M_{\text{yaoyao-peer1}} = 28$, range = 18–31; $M_{\text{yaoyao-peer2}} = 25.2$, range = 12–32; $M_{\text{yaoyao-peer3}} = 39$, range = 38–40, Tau-$U = .96$; $p = .00$) increased rapidly to a much higher level in comparison with baseline. Similarly, Baobao's social interactions increased promptly (social initiation: $M_{\text{baobao-peer4}} = 29.3$, range = 24–33; $M_{\text{baobao-peer5}} = 26.2$, range = 22–33; $M_{\text{baobao-peer6}} = 33.6$, range = 31–36, Tau-$U = .93$; $p = .00$; social response: $M_{\text{baobao-peer4}} = 36.3$, range = 34–38; $M_{\text{baobao-peer5}} = 32.8$, range = 3–35; $M_{\text{baobao-peer6}} = 43.2$, range = 39–46, Tau-$U = .92$; $p = .00$). Meanwhile, Yaoyao's inappropriate social interactions showed a slight increase from Baseline 2, but with an overall progressive decreasing trend during the Intervention 2 condition ($M_{\text{yaoyao-peer1}} = 1.67$, range = 0–4; $M_{\text{yaoyao-peer2}} = 3$, range = 1–4; $M_{\text{yaoyao-peer3}} = 1.2$, range = 0–2). Baobao's inappropriate social interactions decreased slightly during Intervention 2 sessions compared with Baseline 2 conditions ($M_{\text{baobao-peer4}} = 0.17$, range = 0–1; $M_{\text{baobao-peer5}} = 0.2$, range = 0–1; $M_{\text{baobao-peer6}} = 0$, range = 0–0). During follow-up sessions, Yaoyao demonstrated only occasional occurrences of inappropriate social interactions, and Baobao did not show any instances of inappropriate social interactions.

Typically Developing Peers

Figures 3 and 4 depict the social initiations, social responses, and inappropriate social interactions of the typically developing peers. During the first baseline, most peers initiated and responded to Yaoyao and Baobao, though these behaviors were at low levels. With the implementation of Intervention 1, along with increases in social initiations and responses of the child with ASD, the level of the social initiations and responses from typically developing peers increased markedly (Tau-$U = .85$; $p = .00$ for Yaoyao, Tau-$U = .93$, $p = .00$ for Baobao) during this phase.

With the introduction of the second baseline, peer interaction again decreased to low or zero levels. With the implementation of Intervention 2, the peers' social interaction behaviors increased quite significantly from baseline (Tau-$U = .92$; $p = .00$ for Yaoyao's peers and Tau-$U = .94$; $p = .00$ for Baobao's peers). Moreover, Intervention 2 for Yaoyao's peers shows a clear ascending stable trend for Peer 1 and 2. Peer 1 showed one fluctuation then reached a stable ascending level at later sessions. Baobao's peers also showed a similar level and trend in Intervention 2.

The inappropriate social interactions for all peers occurred at low levels across all conditions. Yaoyao's peers displayed a few instances of inappropriate social interactions during baseline, which increased slightly during both Intervention 1 and 2 and maintenance sessions. Baobao's peers exhibited low levels of inappropriate social interactions both in intervention and baseline conditions.

Engagement in Social Play

Figures 5 and 6 depict engagement in social play for Yaoyao and Baobao and their peer partners during the 15-min sessions. During the first and second baseline, both participants had a low level of engagement in social play with peers. As Intervention 1 was introduced, participants' time engaged in social play with peers immediately increased to a relatively high and stable level (Tau-$U = .90$; $p = .00$ for Yaoyao; Tau-$U = .90$; $p = .00$ for Baobao). The same pattern occurred with Intervention 2 (Tau-$U = .96$; $p = .00$ for Yaoyao; Tau-$U = .92$; $p = .00$ for Baobao).

Social Validity

Teachers and parents found the intervention procedures to be acceptable ($M = 5$; $SD = 0$ for teachers, $M = 5$; $SD = 0$ for parents), effective ($M = 4.6$; $SD = 0.52$ for teachers, $M = 5$; $SD = 0$ for parents), and satisfactory ($M = 4.8$; $SD = 0.39$ for teachers, $M = 5$; $SD = 0$ for parents). Peers reported that Yaoyao became more "talkative" and they expressed willingness to play with her more often since they had "much better interest in her." Moreover, Baobao's parents reported him to play and talk more with them at home, and Baobao expressed his enjoyment playing with peers in class to his parents.

Discussion

This study was the first attempt to investigate the effects of a preferred activity with peer support intervention package on the social behavior of preschoolers with ASD and their typically developing peers in inclusive Chinese classrooms. Both intervention conditions were successful in increasing initiations, responses, and social play in participants with

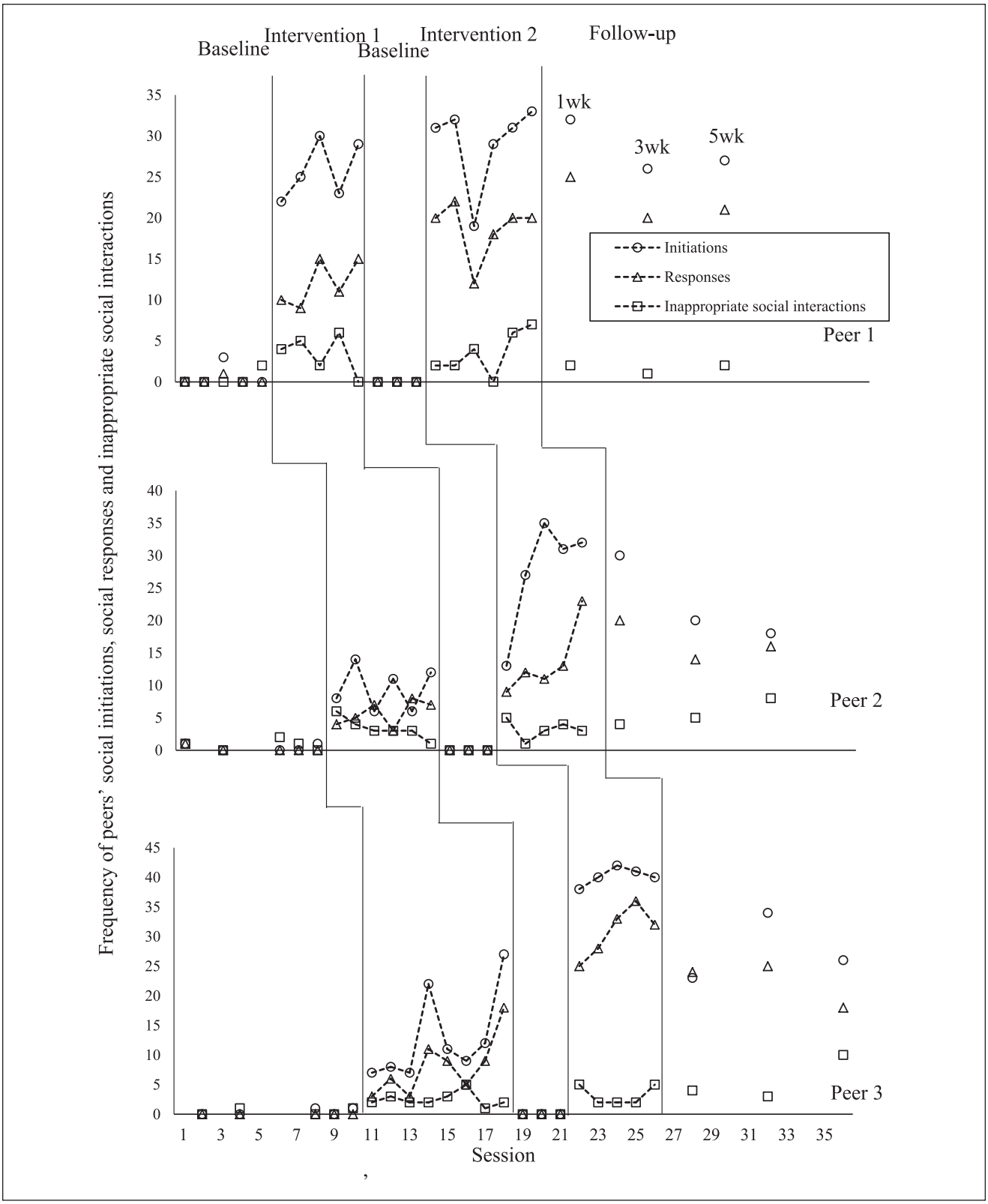


Figure 3. Frequency of peers' social initiations, social responses, and inappropriate social interactions directed to Yaoyao per play session.

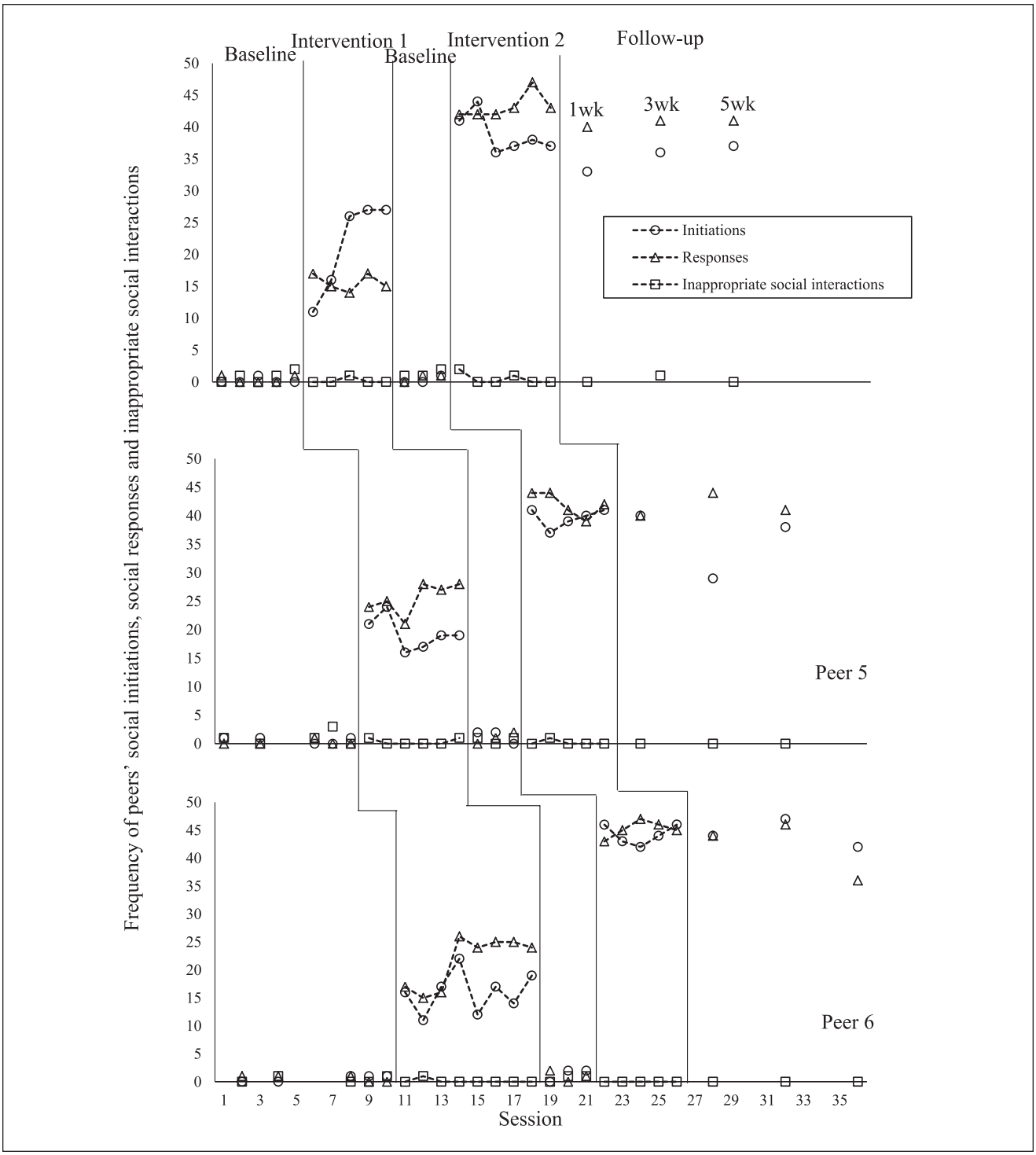


Figure 4. Frequency of peers' social initiations, social responses, and inappropriate behaviors directed to Baobao per play session.

ASD. Maintenance was demonstrated in the preferred activity plus peer support condition. Although participants demonstrated the most social interactions in the preferred activities plus peer support condition, design limitations prevent us from offering conclusive findings on the effects of peer support strategies as order effects may have influenced results and a peer support alone condition was not assessed; results must thus be interpreted cautiously. The intervention, however, resulted in a clinically significant change in social behavior for participants and peers

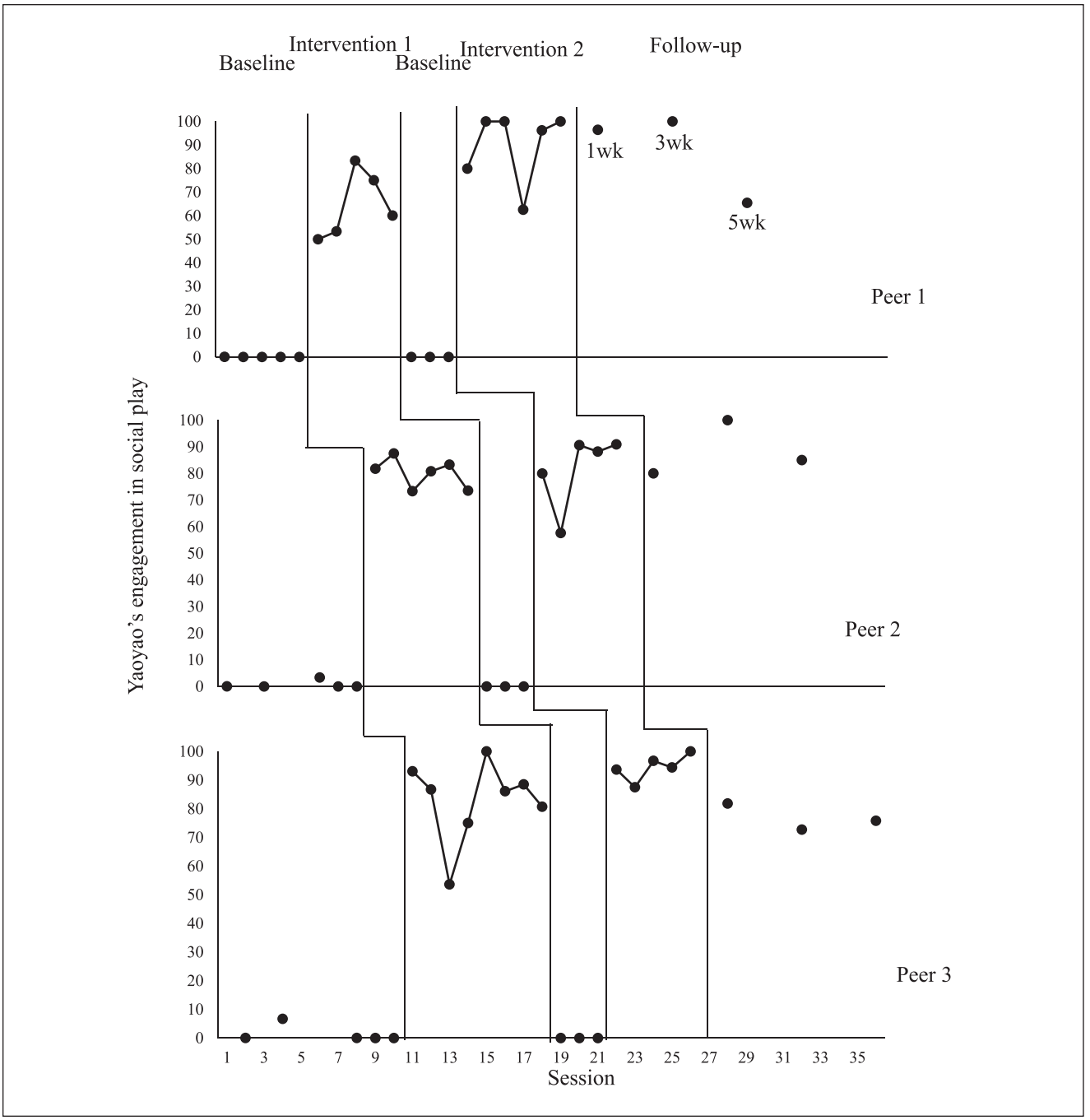


Figure 5. Yaoyao's engagement in social play.

and demonstrated a high level of social validity within an inclusive Chinese preschool setting.

Children With ASD

These findings support previous research demonstrating that social skill interventions incorporating the preferred interests and activities of children with ASD (e.g., Watkins, O'Reilly, et al., 2019) with peer support strategies (e.g., Hu et al., 2018) can improve the social interaction behaviors of both preschoolers with ASD and their typically developing peers in inclusive settings. In addition, positive maintenance results were observed in the peer support condition, which may be attributed to the use of peer

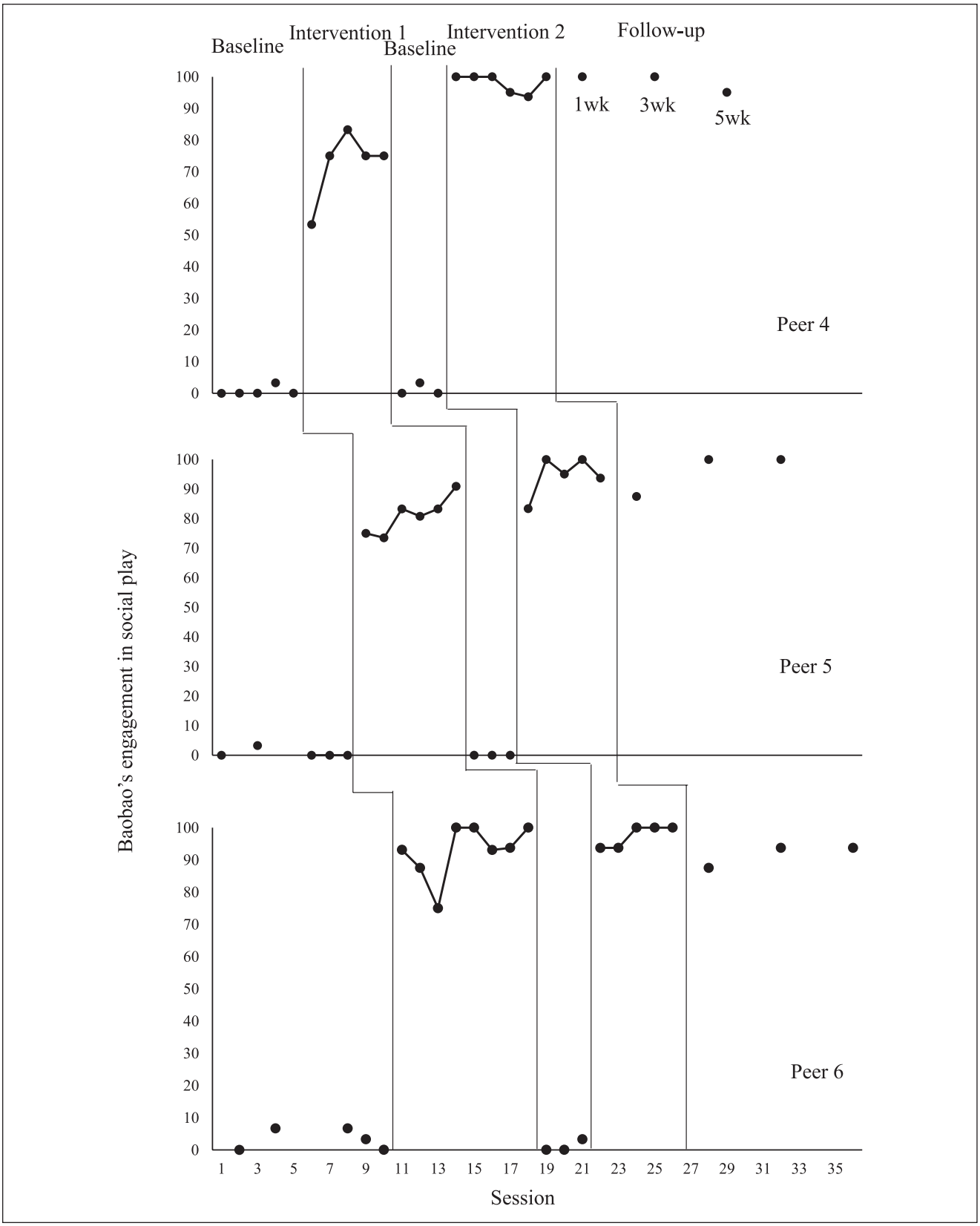


Figure 6. Baobao's engagement in social play.

training strategies that have been shown to contribute to skill maintenance (Watkins et al., 2015). Furthermore, the positive social validity findings support the acceptability of the intervention procedures and outcomes within an inclusive Chinese preschool context. It should be noted that with only two participants with ASD, these findings are limited and future replications are needed to support this approach. Still, this study has important implications for inclusive education and demonstrates that preferred activities and peer support may effectively facilitate and maintain interaction among children with ASD and their typically developing peers.

The preferred activity procedures in isolation (i.e., Intervention 1) were successful in increasing interaction and engagement in social play between the child with ASD and typically developing peers compared with baseline play sessions. This supports previous research that has demonstrated that incorporating preferred activities of the child with ASD into activities, along with behavioral strategies such as modeling, can improve socialization with peers (e.g., Watkins, O'Reilly, et al., 2019). It is plausible that the use of preferred activities increases the motivation of the child with ASD to interact and participate in social activities and play, thus resulting in increased social behavior (Koegel & Koegel, 2006). Embedding a child's preferred activities into play with typically developing classmates and modeling are fairly simple intervention strategies, which would seem a feasible approach for use in an inclusive classroom by teachers who may not have specialized training in behavioral interventions. It is important to note, however, that social interaction decreased when the preferred activity intervention was withdrawn, which has also been demonstrated in previous studies using preferred interests and activities (e.g., Koegel et al., 2013; Watkins, O'Reilly, et al., 2019). Thus, additional strategies may likely be needed to help support and maintain these interactions.

The addition of peer support during Intervention 2 likely resulted in high levels of social interaction as peers were taught to directly reinforce the social behaviors of the children with ASD, thus further promoting their acquisition and use of these skills. Peer support, when used in combination with other strategies, may possibly produce a greater combined effect than strategies used in isolation, though this must be interpreted cautiously as the present study did not directly compare the two intervention conditions. However, these results contribute to the large body of research demonstrating the effectiveness of peer training strategies and further support previous research using peer support to increase interaction in preferred play activities (Hu et al., 2018). Future research is needed that compares peer support strategies within the context of preferred activities to peer support during nonpreferred or less highly preferred play activities to examine the generalization of these findings.

The successful maintenance of results also reflects previous findings suggesting that peer characteristics similar to what we used in the selection criteria for this study (i.e., typical language and social skills, prior interest in interacting with children with ASD, willingness to serve as a peer, and compliance with teacher directions) are likely to yield positive maintenance effects (Watkins et al., 2015). These peer characteristics are important considerations for teachers when selecting typically developing classmates to participate in an intervention.

Typically Developing Children

Moreover, this study demonstrated the effectiveness of both strategies in increasing the frequency of social interactions of typically developing peers, although additional work is needed to directly compare any differential effects between strategies. Furthermore, peers' social behaviors increased in the combined preferred activity plus peer support condition and follow-up sessions. The typically developing peers learned how to provide prompts and use positive reinforcement with participants, and importantly, peers were also trained in strategies to help maintain social interaction if the child with ASD did not respond to their initial attempts, which likely led to significant increases in the peers' social interactions. Our results also further demonstrate the utility of peer support strategies in facilitating the maintenance of social skills for children with ASD, as well as demonstrating that peer support can be incorporated into the context of daily activities in inclusive preschool classrooms and provide preschoolers with ASD with opportunities to interact with a variety of peers to promote social interactions (McGrath et al., 2003). Notably, peers also reported positive feedback on the intervention. The acceptability and impact of such procedures on peers are an important consideration when implementing interventions for children with ASD involving their classmates (DiSalvo & Oswald, 2002).

Inappropriate Social Interactions

Inappropriate social interactions were evident to some degree for the participants and typically developing peers across conditions. For Yaoyao and her peers, inappropriate social interactions only occurred sporadically in baseline sessions, likely due to the fact that the participants and peers showed limited engagement and interaction with each other. Within the intervention conditions, Yaoyao and her peers' inappropriate social interactions were slightly elevated compared with baseline conditions, likely because there was an increased opportunity to engage in such behaviors. For Baobao and his peers, inappropriate social interactions remained low and stable throughout baseline and intervention.

For children with ASD who exhibit high levels of inappropriate or challenging behavior in the context of interactive play activities, additional strategies may be needed to directly address the reduction of inappropriate behaviors in addition to strategies that support and maintain social interactions. This seems especially needed within an inclusive classroom context as challenging behavior of children with ASD has been shown to be one of the biggest barriers to successful inclusion (Crosland & Dunlap, 2012).

Limitations and Implications

A few limitations of this study must be addressed. As previously noted, we did not assess the peer support condition in isolation outside of the context of the preferred activity, which is a design limitation of this study and prevents us from making direct comparisons between the results of the intervention conditions. It is possible that order or sequence effects contributed to the positive results in the second intervention condition as the preferred activity alone condition may have primed the children's social behavior when the preferred activity with peer support was reintroduced. Thus, findings must be interpreted judiciously. Future research might use an alternating treatment design to directly compare intervention conditions on the social interactions of preschoolers with ASD and rule out potential sequence effects. Research is also needed to establish the generality of effects with multiple and varied participants and peers, more activities in daily routines, and across settings. Future studies should also determine the amount of time peer-supported strategies need to be included in play activities to reach a high and stable level of social interaction and how to fade peer support without decreases in social interaction. In addition, training teachers to implement and monitor these interventions is another important direction for future research. In this study, researchers rather than teachers were responsible for peer training and implementing and monitoring the intervention, so the feasibility of this approach when implemented by teachers should be assessed in future studies. Despite these limitations, this study provides further evidence of the efficacy of both peer support strategies and embedding preferred activities of children with ASD within play interventions. This study has implications for inclusive education across varied cultural contexts and demonstrates that preferred activities and peer-mediation may effectively facilitate interaction among children with ASD and typically developing peers.

Compliance With Ethical Standards

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Informed Consent

Informed consent was obtained from all teachers and school administrators included in the study. Informed consent was also obtained from the participants' parents/legal tutors.

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